

ESP 32 BASED WATER LEVEL INDICATOR WITH AUDIBLE ALARM SYSTEM

BY :

G.AYYAPPA ASHISH KUMAR (23EEB0B21)

CH. ROHIT SATYA MANIKANTA (23EEB0B22)

PINTU KUMAR (23EEB0B52)

Background and Significance of the Project

Background:

In day-to-day life, water tanks in homes, schools, and industries are often filled manually without knowing the exact water level. This can lead to **overflow**, **wastage of water**, or **pump damage** when the tank runs dry. Traditional methods such as using floats or manual checking are inconvenient and less accurate.

With the availability of modern microcontrollers like the **ESP32**, it is possible to design an automatic system that **detects the water level accurately** and gives **visual and audible indications**.

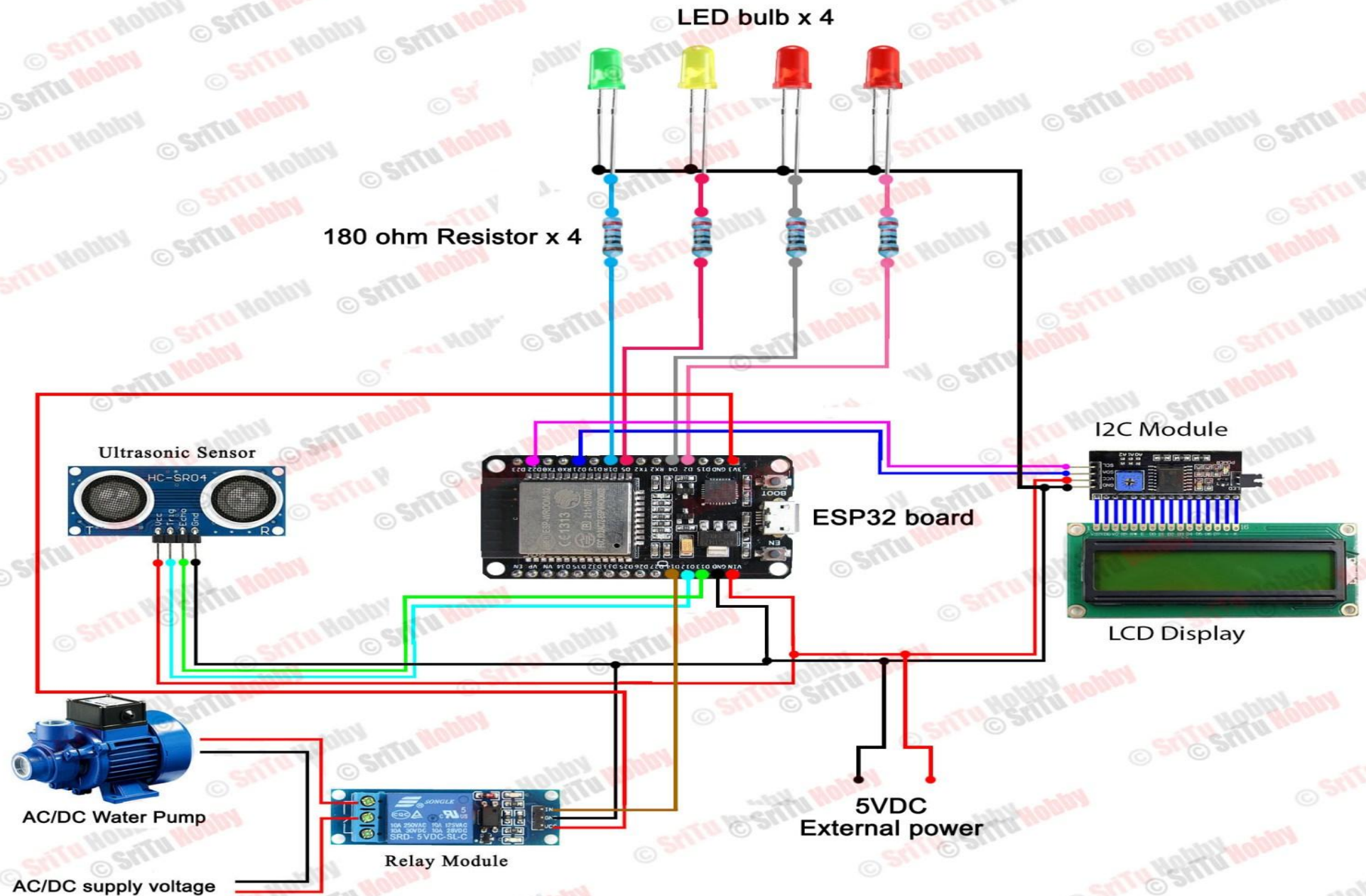
Significance:

- Helps in **preventing water wastage** due to overflow.
- **Provides clear indication** of water levels using LEDs and a buzzer.
- **Eliminates the need for manual checking** of water levels.
- **Low-cost and reliable** system suitable for domestic and industrial use.
- Ensures **longer pump life** by preventing dry running.

Components used Interfacing

Component	ESP32 Pin	Function
Ultrasonic Sensor Trig	GPIO 12	Trigger Signal
Ultrasonic Sensor Echo	GPIO 13	Echo Input
LED (Low Level)	GPIO 05	Indicates Low Water
RELAY PIN	GPIO 14	Acts as switch for motor control
LED (Full Level)	GPIO 04	Indicates Full Tank
Buzzer	GPIO 18	Audible Alarm
VCC & GND	5V & GND	Power Supply

CIRCUIT DIAGRAM



Logic and programming

Basic Logic:

- The **ultrasonic sensor** continuously measures the distance from the top of the tank to the water surface.
- The **ESP32** calculates the **water level** by comparing the measured distance with the total height of the tank.
- Depending on the level, specific **LEDs** are turned ON.
- When the water reaches the **maximum level**, the **buzzer** is activated to give an alarm.
- When the water is at a **low level**, the **low-level LED** glows and the buzzer may also sound if desired.

Step-by-Step Flow:

1. Initialize all pins (sensor, LEDs, buzzer).
2. Send trigger pulse to ultrasonic sensor.
3. Receive echo and calculate distance.
4. Convert distance to water level percentage.
5. Use `if-else` conditions to:
 1. Turn ON/OFF LEDs based on level.
 2. Activate buzzer for full or low levels.
6. Continuously repeat the process in a loop.

Observations, Outcomes & Future scope

Observations

- The ultrasonic sensor accurately detects the water level in the tank.
- LEDs change according to the water level (Low → Normal → Full).
- The buzzer gives an audible alert when the tank is **full** or **almost empty**.
- The ESP32 performs reliable and quick switching between levels.
- System consumes very little power and works efficiently on 5V supply.

Outcomes

- Successfully developed a **low-cost automatic water level indicator**.
- The system effectively **reduces water wastage** by alerting when the tank is full.
- Provides **clear visual and sound indications** for different levels.
- **Reduces manual effort** in checking and controlling water levels.
- Compact and can be easily installed for **domestic and industrial tanks**.

Future Scope

- Add **automatic motor control** (pump ON/OFF based on level).
- Implement **LCD or OLED display** for showing percentage of water level.
- Use **Bluetooth/Wi-Fi module** for mobile notifications (optional future IoT upgrade).
- Design a **PCB version** for long-term stable use.
- Apply for **large water tanks** or **underground reservoirs** with minimal modification

Our Learnings

- We learned how to **interface sensors and output devices** such as ultrasonic sensors, LEDs, and buzzers with the ESP32 microcontroller.
- We understood the **working principle of ultrasonic distance measurement** and its application in level detection.
- We gained hands-on experience in **programming the ESP32** using the Arduino IDE.
- We learned to apply **conditional logic and control techniques** for real-time systems.
- We improved our **hardware assembling and debugging skills** during testing.
- We realized the importance of **automation in water management** and **resource conservation**.

Conclusion

- The project “**ESP32 Based Water Level Indicator with Audible Alarm System**” was successfully designed and implemented.
- The system **accurately monitors water levels** and provides both **visual (LED)** and **audible (buzzer)** indications.
- It effectively **reduces water wastage** and **manual monitoring efforts**.
- The design is **cost-effective, reliable, and suitable** for homes, schools, and industries.
- This project helped us gain practical knowledge of **embedded systems and sensor applications**

Individual Contribution

Ch.Rohit (Hardware & Circuit Designer)

“I was responsible for the hardware interfacing part of the project. I connected and tested all components — ultrasonic sensor, LEDs, relay module, and buzzer through BC547 transistor — with the ESP32. I verified the power supply and connections and ensured the motor and relay worked properly.”

Ashish Kumar (Coding & Logic Developer)

“I worked on the software and programming part of the project. I developed and tested the Arduino code for ESP32, including logic for ultrasonic sensor readings, threshold detection, LED/buzzer indication, and motor control. I also integrated the Wi-Fi and Blynk IoT connectivity features.”

Pintu Kumar (LCD Display, Blynk IoT, Report & Presentation)

“I handled the IoT monitoring, LCD interfacing, and documentation part. I set up the Blynk dashboard, linked virtual pins (V0 and V1), and configured Wi-Fi settings. I also prepared the presentation slides, circuit diagram, and assisted in testing the display output and demo.”



Thank You